

FalconVerifier

Falcon says it. Lean 4 proves it — or refutes it. In Arabic.

A verification middleware that turns Falcon's Arabic math and logic answers into kernel-checked claims, teaches Falcon from what the kernel refutes, and keeps an auditable trace of every step.

Falcon is the region's model. Its users write Arabic. Its training data mostly did not.

- Falcon-7B/40B/180B: 3.5T tokens, **75% English** RefinedWeb; Arabic not among the listed languages.
- Falcon-H1: ~11T English web tokens; Arabic is one of 17 languages sharing a 3,000 GT pool.
- Falcon-H1-Arabic: ~100B Arabic tokens of continued pre-training — under 1% of what the base saw.

What that looks like

سعر ٨٠٠ درهم خُفِّضَ ١٠٪ ثم ٢٠٪. السعر النهائي؟

A price of 800 AED is reduced 10% then 20%. Final price?

Falcon 3B (first try, benchmark): **680** — it added the discounts. Correct: **576**.

Baseline first-try accuracy on our 244 Arabic problems: **48%**.

The truth of a math claim does not depend on the language it was written in.

So don't ask a second LLM whether Falcon is right. Translate each sentence of Falcon's reasoning into a Lean 4 proposition and let the proof kernel decide.

VERIFIED

The kernel proved P . Not "confident" — proved.

REFUTED

The kernel proved $\neg P$. Falcon gets told, in Arabic, what is false and why.

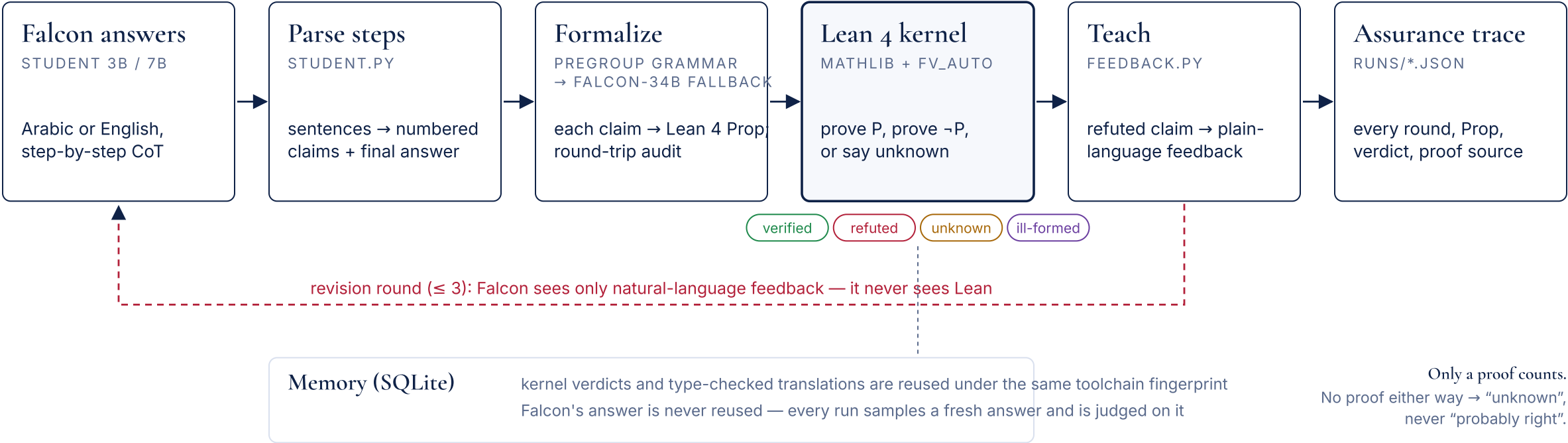
UNKNOWN

No proof either way. We say so. Never "probably right".

One loop, six stages

How a Falcon answer becomes a kernel-checked claim

FALCON SAYS IT · LEAN 4 DECIDES IT · FALCON LEARNS FROM IT



For the common fragments, no LLM writes the Lean at all

A **pregroup grammar** (Lambek) gives every Arabic word a type; a sentence is well-formed when the types cancel. The same derivation is a functor into meaning — so the translation is a *proof*, not a guess.

```
kataba : s · ol · πl      (verb, VS0)
ahmad  : π                (subject)
aḷDarsa: o                (object)
s · ol · πl · π · o → s    ✓
```

Fragments covered: arithmetic, quantifier logic, symmetric relations & counting. Everything else goes to Falcon-H1-Arabic-34B with a round-trip faithfulness audit — a mismatch downgrades to *unknown*.

Five friends

خمسة طلاب يجلسون في الفصل، ويقول كل واحد منهم
إن ثلاثة من الأربعة الباقين أصدقاؤه. هل يلزم أن أحدهم
يكذب؟

Five students sit in class; each says three of the other four are his friends. Must one of them be lying?

```
∀ f : Fin 5 → Fin 5 → Bool, ¬ Regular f 3
```

Lean proves it via the handshake lemma (odd degree sum) — no enumeration, and nobody had to understand the word "friend". VERIFIED in 18 s.

Two examples, one screen

1 · Catch and correct

هاتف بـ 800 درهم، خُفِّضَ 10% ثم 20%

A phone at 800 AED, -10% then -20%

Falcon 3B answers → step 3 turns **REFUTED** with the kernel's $(800:\mathbb{R}) \cdot 0.9 \cdot 0.8 = 576 \rightarrow$ Arabic feedback → round 2 **VERIFIED**, assurance 1.0.

If 3B happens to be right first time: every step is still green with its proof — that *is* the product.

2 · Prove without understanding

خمسة طلاب، كل واحد له ثلاثة أصدقاء

Five students, each with three friends

Deterministic grammar → handshake theorem → answer «نعم» (yes, someone lies) **VERIFIED**. Open "Lean 4 source (audit)": that is the exact file the kernel checked. Ask it again: the kernel verdict comes from memory in 0 s — Falcon's answer is still fresh.

Falcon 3B, Arabic, Falcon-34B as formalizer — Falcon verifies Falcon

48% → 76%
ACCURACY, 244 PROBLEMS

92%
WRONG ANSWERS CAUGHT BY
LEAN

0
CORRECT ANSWERS MADE WRONG

1
FALSE ALARM IN 244

SET	N	BASELINE	VERIFIED	CAUGHT	FIXED	REGRESSIONS
Arabic curated	86	58%	88%	33/36	26/36	0
Arabic scale	244	48%	76%	117/127	69/127	0

Every row links to its problems, traces and Lean sources on the /benchmark page. 294 Lean-labelled preference pairs already exported for DPO.

Middleware, not a model

For any Falcon app

```
POST /api/solve
{ "problem": "...", "rounds": 3 }
→ verdicts, feedback, trace
```

One call adds a kernel-backed assurance score to Falcon output.

For TII

Every refutation is a training signal labelled by a proof, not by a human or an LLM judge. The flywheel is already a file.

Open & deployable

Docker image with Lean + Mathlib, HTTPS, rate limits, memory. Web UI, CLI, API — all in the repo.

Limits we state out loud: Lean checks what was formalized, not the prose; out-of-fragment steps depend on the 34B formalizer; proof-style problems are "unknown" today.

Trust in Arabic, proved in Lean.

A regional model has to lead in its own language. We give Falcon a kernel that never bluffs — and a way to learn from it.

github.com/hw4375-collab/FalconVerifier · live demo on the next screen